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Introducing soft TopK quadratic convolution to improve the performance of neural networks in medical image segmentation

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ABSTRACT

Medical image segmentation is crucial for computer-aided diagnosis and treatment planning, where accurate organ delineation serves as the foundation for reliable quantitative analysis. Standard convolutional layers primarily extract local features through linear projections within the receptive field, which to some extent limits their ability to model complex nonlinear local responses. To address this, we propose a Soft TopK Quadratic Convolution (STKQC) module as a plug-in enhancement unit for standard convolutional layers. STKQC introduces a parallel quadratic feature modeling branch alongside the traditional linear convolution branch. Unlike explicitly constructing a complete quadratic matrix over the entire local patch, STKQC employs soft-pooling-style neighborhood feature selection through learnable local scale selection and dynamic grid adjustment, and further constructs a lightweight diagonal quadratic response. This approach introduces local second-order nonlinear information while effectively controlling additional parameters and computational overhead. We evaluated STKQC on the MICCAI 2017 cardiac biventricular segmentation dataset and the AMOS22 abdominal multi-organ segmentation dataset. Experimental results demonstrate that integrating STKQC into a typical encoder-decoder network improves model performance in segmentation tasks, validating its effectiveness as an efficient convolutional enhancement module for medical image segmentation.

1. Introduction

Medical Image Segmentation is of critical importance in the medical field (Prevedello et al., 2019). Precise segmentation of organs or lesion areas in images helps doctors diagnose diseases more accurately. When determining surgical plans, it provides precise anatomical structure information to optimize treatment plans. Alternatively, during radiation therapy for patients, it enables precise delineation of target areas, allowing radiation therapy plans to be designed with greater precision and reducing radiation damage to critical organs. By comparing segmentation results at different time points before and after treatment, treatment efficacy and disease progression can be assessed. However, medical image segmentation still faces significant challenges. First, some images have poor image quality, with issues such as noise, artifacts, and low contrast increasing segmentation difficulty. Second, unlike other segmentation tasks, organs and lesions vary greatly in shape, size, and position, and blurred boundaries further reduce segmentation accuracy. Imaging data from different hospitals is not interoperable and is not publicly available, making it difficult to obtain high-quality annotated data, and the annotation process is time-consuming and labor-intensive. High-resolution image processing requires significant computational resources, especially for 3D images. In clinical applications, algorithms must deliver results quickly while maintaining accuracy. Data for certain diseases is scarce, necessitating high-precision segmentation with small sample sizes. Therefore, medical image segmentation is indispensable in precision medicine but must overcome multiple challenges such as image quality, data annotation, and algorithm generalization. In the future, as technology advances, these challenges are expected to be gradually resolved, further enhancing medical diagnosis and treatment standards.

Inspired by the convolutional neural networks (He et al., 2016; Krizhevsky et al., 2012) used in ImagNet (Deng et al., 2009), various convolutional network architectures have emerged in the field of image segmentation, such as the U-Net (Ronneberger et al., 2015) encoder-decoder architecture (Shelhamer et al., 2017), the Transformer (Vaswani et al., 2017), and the generative adversarial architecture (Goodfellow et al., 2014). However, these classic model architectures

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still use classic first-order convolutions, i.e., linear convolutions. While linear convolutions enable models to extract linear features at each layer, they fail to capture higher-dimensional features present in the data.

Some scholars have also explored how to utilize the second-order features of data (Gao et al., 2019; Li et al., 2017), but most of them use attention modules nested in the network to assist in feature extraction. In order to enable the network to apply convolution to extract higher-dimensional data features at each layer, accelerate model convergence, and improve model inference performance, there is an urgent need for a convolution method that can identify higher-dimensional data features.

In this paper, we propose a Soft TopK Quadratic Convolution (STKQC) module as a plug-in extension of standard convolution. Instead of constructing a dense quadratic matrix over the entire local receptive field, STKQC introduces a soft-pooling-guided diagonal quadratic branch in parallel with the conventional linear convolution branch. This design enables the convolutional operator to incorporate local second-order nonlinear responses while preserving dense spatial outputs and maintaining a controllable parameter cost. In this study, STKQC is evaluated in medical image segmentation tasks by integrating it into representative encoder–decoder networks. Experimental results show that the proposed module can improve segmentation performance under the same training settings, suggesting its potential as a lightweight convolutional enhancement module.

The remainder of this manuscript is organized as follows. Section 1 introduces the background and motivation of convolutional neural networks for visual recognition and medical image segmentation. Section 2 reviews related work on medical image segmentation networks and second-order feature modeling. Section 3 presents the proposed STKQC module, including its mathematical formulation, relation to existing second-order methods, backpropagation, and complexity analysis. Section 4 describes the datasets, experimental settings, evaluation metrics, and results. Section 5 concludes the paper and discusses future work.

2. Related works

In computer vision, the task of image segmentation is to divide an image into multiple regions or objects. Segmentation provides a pixel-level understanding of the image. In the field of medical image analysis, segmentation is used to accurately identify human tissues and organs and locate objects to assist doctors in diagnosis. To this end, many scholars and research institutes have attempted different types of methods, such as traditional machine learning methods and, in the era of artificial intelligence, brand-new deep neural network methods (LeCun et al., 2015).

Traditional machine learning methods such as random forests (Pereira et al., 2016), support vector machines (Zheng et al., 2018), k-nearest neighbors (Steenwijk et al., 2013), graph cuts (Boykov & Jolly, 2001), watershed (Grau et al., 2004), active contour models (Zhang et al., 2020), and level sets (Liu et al., 2016) all require pre-designed features such as color, texture, and edge extraction. Selecting the most discriminative features to reduce computational complexity has become a major constraint on the development of such methods.

Unlike machine learning methods, deep neural networks can automatically learn and extract multi-level features. Through end-to-end learning, the algorithm can directly learn complex patterns and structures from raw data, simplifying the feature extraction process. Additionally, through backpropagation algorithms and gradient descent optimization methods, the network can automatically adjust training parameters. During training, combining appropriate loss functions accelerates model convergence and improves model inference performance. These automated and adaptive characteristics enable deep neural networks to excel in handling high-dimensional, nonlinear, and large-scale data, particularly in complex tasks such as medical image segmentation, where they achieve higher accuracy and robustness. Algorithms based on deep neural networks significantly enhance the efficiency and effectiveness of image segmentation tasks.

The emergence of transposed convolutions (Shelhamer et al., 2017) has driven the development of end-to-end encoder-decoder architectures. The encoder uses convolution and pooling operations to convert high-resolution images into low-resolution feature representations that retain deep semantic information; the decoder then progressively upsamples and restores these low-resolution feature maps to reconstruct high-resolution output images, enabling pixel-level precise segmentation. The encoder-decoder architecture, as a symmetric structure, enables the network to capture rich semantic information while preserving spatial information. Therefore, networks based on this architecture (Abdeltawab et al., 2020; Jethi et al., 2020; Khened et al., 2018) are also widely applied in medical image analysis.

Unlike traditional convolutional neural networks (CNNs), Transformers utilize self-attention mechanisms to capture long-range dependencies in images, thereby better understanding global contextual information. As a result, Transformers are capable of processing complex scenes and images rich in detail. Models such as Vision Transformer (ViT) (Dosovitskiy et al., 2021) and Swin Transformer (Liu et al., 2021) divide images into multiple small patches, flatten them into sequences, use the Transformer encoder for feature extraction, and then reconstruct high-resolution segmentation results via the decoder. Networks improved based on this architecture (Cai et al., 2022; Cao et al., 2023; Grzeszczyk et al., 2022) have also been applied in the field of medical image analysis. Although the Transformer shows great potential in image segmentation, its dependence on large-scale data and high computational resource requirements remain challenges that need to be overcome.

Generative Adversarial Networks (GANs) consist of two components: a generator and a discriminator. The generator is responsible for generating realistic segmentation results, while the discriminator distinguishes between the generated results and the true labels. This adversarial training approach drives the generator to continuously optimize and produce high-quality segmentation outputs. GANs perform exceptionally well in handling complex textures and edge regions and exhibit strong robustness when faced with noise and incomplete data. Improved GAN-based methods (Frid-Adar et al., 2018; Han et al., 2018) also demonstrate advantages in processing complex medical images. GANs offer new insights and approaches for image segmentation tasks, and despite some challenges, their potential for enhancing segmentation accuracy and detail restoration cannot be overlooked.

Despite the large number of medical image networks based on the above methods, whether supervised learning, semi-supervised learning (Liu et al., 2025; Wang et al., 2025; Zhang et al., 2025), or large model-based methods, the vast majority of networks still use primary convolution operations to extract linear features from the data, rather than extracting higher-dimensional data features based on convolution (see Fig. 1).

3. Methods

3.1. Traditional linear convolution and quadratic convolutions

In CNNs, standard convolutional layers typically perform feature extraction through a linearly weighted sum within a local receptive field. Let the input feature map be

$$\mathbf{X} \in \mathbb{R}^{B\times C\times H\times W}, \tag{1}$$

Here, B denotes the batch size, C denotes the number of input channels, and H and W denote the height and width of the feature map, respectively. For an output channel o and spatial location (h, w), a standard 2D convolution can be expressed as

$$Y^{\text{lin}}_{b,o,h,w} = \sum_{c=1}^{C}\sum_{u=1}^{k_h}\sum_{v=1}^{k_w} W^{\text{lin}}_{o,c,u,v} X_{b,c,h+u,w+v} + b_o, \tag{2}$$

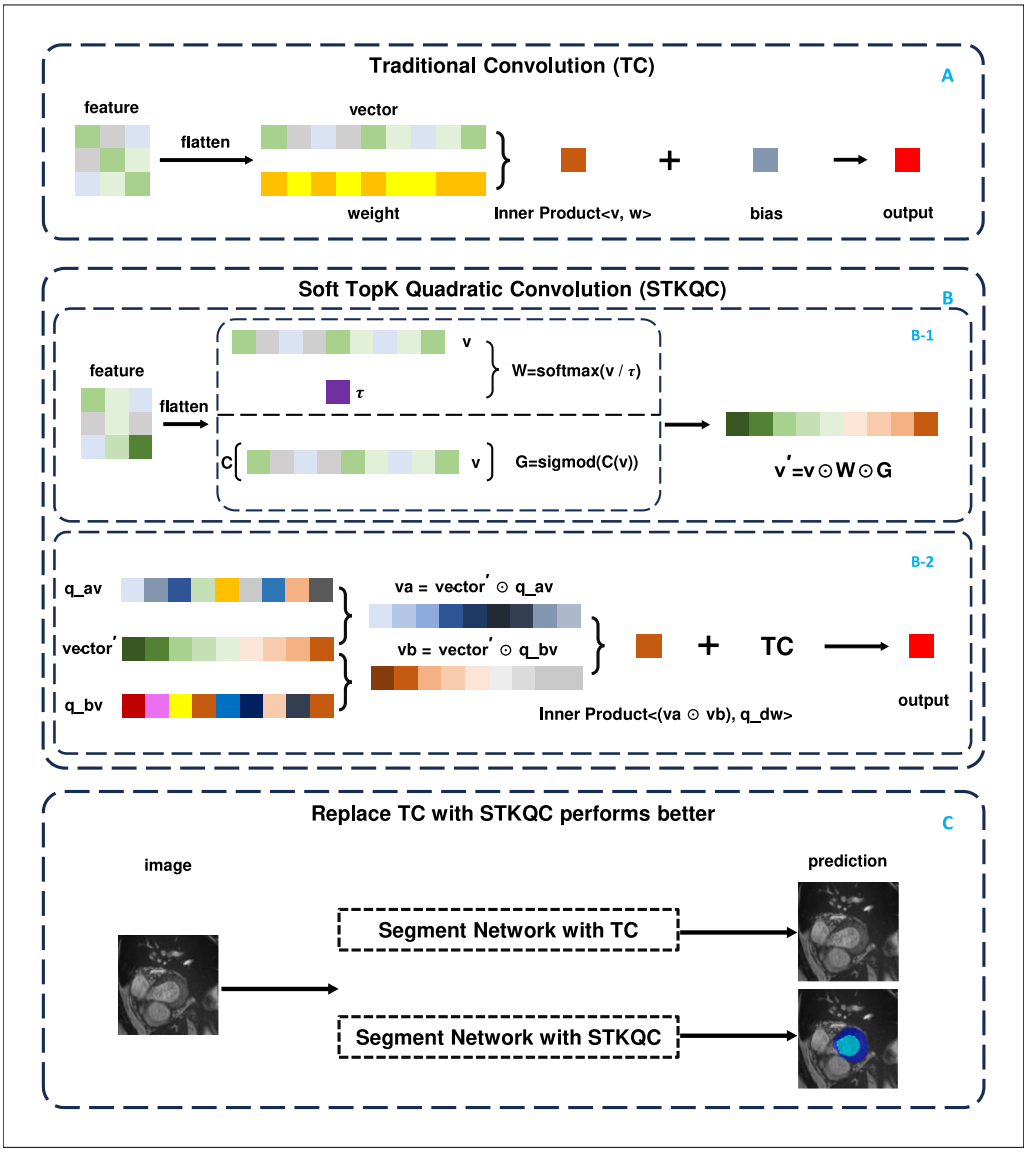


Fig. 1. This figure illustrates the computational process of STKQC. (A) shows the computational process of traditional convolution; (B) illustrates the computational method of STKQC, where (B-1) demonstrates that the data is first reweighted before computing the quadratic terms; (B-2) shows that this method computes the quadratic terms of the data simultaneously with traditional convolution; (C) indicates that introducing STKQC can effectively enhance the model's inference capabilities.

Here, W^{lin} denotes the standard convolution kernel parameters, b_o is the bias term, and $k_h \times k_w$ represents the size of the convolution kernel.

To simplify the notation, the local window corresponding to each spatial location can be expanded into a vector:

$$\mathbf{x}_{b,h,w} \in \mathbb{R}^{CK}, \qquad K = k_h \times k_w. \tag{3}$$

Therefore, standard convolution can be written in the form of a linear mapping:

$$\mathbf{y}^{\text{lin}}_{b,h,w} = \mathbf{W}^{\text{lin}}\mathbf{x}_{b,h,w} + \mathbf{b}, \tag{4}$$

where

$$\mathbf{W}^{\text{lin}} \in \mathbb{R}^{O\times CK}. \tag{5}$$

A Standard Convolution is essentially a first-order linear mapping whose output depends primarily on a weighted sum of local features. Although convolutional neural networks can enhance their expressive power through nonlinear activation functions and stacked layers, a single convolutional layer alone has limited ability to capture higher-order feature relationships within a local region.

To enhance the ability to model local nonlinearities, quadratic mappings can be introduced into the convolution process. The general form of a quadratic convolution can be written as

$$y^{\text{quad}}_{b,o,h,w} = \mathbf{x}^{\text{T}}_{b,h,w}\mathbf{A}_o\mathbf{x}_{b,h,w}, \tag{6}$$

where,

$$\mathbf{A}_o \in \mathbb{R}^{CK\times CK} \tag{7}$$

The quadratic form parameter matrix corresponding to the oth output channel. If the linear convolution branch is also retained, the complete quadratic convolution can be expressed as

$$y_{b,o,h,w} = \mathbf{w}^{\text{lin}}_o\mathbf{x}_{b,h,w} + \mathbf{x}^{\text{T}}_{b,h,w}\mathbf{A}_o\mathbf{x}_{b,h,w} + b_o. \tag{8}$$

This formulation allows for the explicit modeling of second-order relationships between different locations and channels within a local window. However, the full quadratic matrix $\mathbf{A}_o$ has a parameter count of

$$O(C^2K^2), \tag{9}$$

As the number of input channels or the size of the convolution kernels increases, the number of model parameters and computational load rise significantly. Therefore, directly using full quadratic convolution in medical image segmentation networks is often computationally expensive.

To reduce the computational complexity of quadratic modeling, this paper proposes Soft TopK Quadratic Convolution (STKQC). Instead of explicitly constructing a full quadratic matrix, this module introduces local second-order nonlinear modeling capabilities with low parameter overhead through a soft top-k selection mechanism and a lightweight diagonal quadratic response.

3.2. The forward propagation for STKQC

STKQC consists of a standard linear-convolutional-integral branch and a lightweight second-order branch. Given an input feature map

$$\mathbf{X} \in \mathbb{R}^{B\times C\times H\times W}, \tag{10}$$

First, obtain the linear branch output using standard convolution:

$$\mathbf{Y}^{\text{lin}} = \text{Conv}(\mathbf{X}). \tag{11}$$

For the second branch, STKQC first uses the unfold operation to extract local window features at each spatial location:

$$\mathbf{P} = \text{Unfold}(\mathbf{X}), \tag{12}$$

where

$$\mathbf{P} \in \mathbb{R}^{B\times C\times K\times S}, \qquad K = k_h \times k_w, \qquad S = H \times W. \tag{13}$$

Here, $P_{b,c,k,s}$ denotes the feature value of the kth sample point in the local window corresponding to the bth sample, the cth channel, and the sth spatial location.

To enable more flexible feature selection within the local window, STKQC introduces a soft top-k selection mechanism. This mechanism consists of two components: an adaptive spatial selection factor τ and a dimensionality selection graph $\mathbf{G}$. The adaptive spatial selection factor τ acts on the local window dimension to adjust the relative response distributions among different spatial sampling points. The corresponding soft top-k weights can be expressed as

$$\boldsymbol{\alpha} = \text{Softmax}\Big(\frac{\mathbf{P}}{\tau}\Big). \tag{14}$$

A smaller τ makes the weight distribution within a local window more concentrated, causing the model to focus more on spatial locations with stronger responses; a larger τ makes the weight distribution smoother, thereby preserving more complete local neighborhood information. Therefore, τ can be viewed as an adaptive factor that controls the intensity of local response filtering in the spatial domain.

On the other hand, the dimension-selection map $\mathbf{G}$ is used to further modulate the unfolded local features. Specifically, the unfolded patch features are rearranged and fed into a 1×1 convolution, and a dimension-selection map is generated via the sigmoid function:

$$\mathbf{G} = \sigma\big(\text{Conv}_{1\times 1}(\text{Reshape}(\mathbf{P}))\big). \tag{15}$$

Subsequently, the original patch features, the spatial weights derived from τ, and the dimensionality selection map are combined to produce the filtered local features:

$$\widetilde{P}_{b,c,k,s} = P_{b,c,k,s} \cdot \alpha_{b,c,k,s} \cdot G_{b,k,s}. \tag{16}$$

In particular, $\boldsymbol{\alpha}$ primarily performs soft filtering at local spatial locations, while $\mathbf{G}$ further performs dynamic adjustment along the feature dimension. Together, they constitute the soft top-k selection process in STKQC, enabling the quadratic branch to focus more on discriminative responses within the local window.

After obtaining the filtered local features $\widetilde{\mathbf{P}}$, STKQC constructs the quadratic response using two learnable mapping parameters:

$$\mathbf{A} \in \mathbb{R}^{C\times K}, \qquad \mathbf{B} \in \mathbb{R}^{C\times K}. \tag{17}$$

In the implementation, $\mathbf{A}$ and $\mathbf{B}$ correspond to the parameters `q_av` and `q_bv`, respectively. For each local feature element, we have

$$U_{b,c,k,s} = A_{c,k}\widetilde{P}_{b,c,k,s}, \tag{18}$$

$$V_{b,c,k,s} = B_{c,k}\widetilde{P}_{b,c,k,s}. \tag{19}$$

Next, the quadratic response is obtained by element-wise multiplication:

$$R_{b,c,k,s} = U_{b,c,k,s} \cdot V_{b,c,k,s}. \tag{20}$$

Substituting $U_{b,c,k,s}$ and $V_{b,c,k,s}$ into the above equation yields:

$$R_{b,c,k,s} = A_{c,k}B_{c,k}\widetilde{P}^2_{b,c,k,s}. \tag{21}$$

Therefore, the quadratic response in STKQC can be viewed as a form of local quadratic modeling in diagonal form. This design avoids the high parameter count and computational complexity associated with a full quadratic matrix, while retaining the expressive power of local quadratic nonlinearity.

Next, the quadratic response is flattened and mapped to the output channel dimension via a learnable projection matrix:

$$Q_{b,o,s} = \sum_{c=1}^{C}\sum_{k=1}^{K} W^q_{o,c,k} R_{b,c,k,s}, \tag{22}$$

where,

$$\mathbf{W}^q \in \mathbb{R}^{O\times CK}. \tag{23}$$

Finally, the output of the quadratic branch is converted back into a spatial feature map and added to the output of the standard linear convolution branch to obtain the final output of STKQC:

$$Y_{b,o,h,w} = Y^{\text{lin}}_{b,o,h,w} + Q_{b,o,h,w}. \tag{24}$$

If the quadratic branch is not enabled, this module degenerates into a standard convolution.

3.3. The backpropagation for STKQC

Let the loss function be

$$\mathcal{L}, \tag{25}$$

and the gradient from the subsequent network layers be

$$\Delta Y_{b,o,s} = \frac{\partial \mathcal{L}}{\partial Y_{b,o,s}}. \tag{26}$$

Since the output of STKQC consists of both standard convolutional branches and quadratic branches, the gradient of the standard convolutional branches can be computed using standard backpropagation. Below, we primarily describe the gradient propagation process for the core quadratic terms in the quadratic branches.

The second-order branch output is

$$Q_{b,o,s} = \sum_{c=1}^{C}\sum_{k=1}^{K} W^q_{o,c,k} R_{b,c,k,s}, \tag{27}$$

where the quadratic response is

$$R_{b,c,k,s} = A_{c,k}B_{c,k}\widetilde{P}^2_{b,c,k,s}. \tag{28}$$

Therefore, the gradient of the projection weight $W^q_{o,c,k}$ is

$$\frac{\partial \mathcal{L}}{\partial W^q_{o,c,k}} = \sum_{b=1}^{B}\sum_{s=1}^{S} \Delta Y_{b,o,s} R_{b,c,k,s}. \tag{29}$$

By the chain rule, the gradient received by the quadratic response $R_{b,c,k,s}$ is

$$\frac{\partial \mathcal{L}}{\partial R_{b,c,k,s}} = \sum_{o=1}^{O} \Delta Y_{b,o,s} W^q_{o,c,k}. \tag{30}$$

Furthermore, the gradients of the quadratic mapping parameters $A_{c,k}$ and $B_{c,k}$ can be written directly as

$$\frac{\partial \mathcal{L}}{\partial A_{c,k}} = \sum_{b=1}^{B}\sum_{s=1}^{S}\left(\sum_{o=1}^{O}\Delta Y_{b,o,s}W^{q}_{o,c,k}\right)B_{c,k}\widetilde{P}^{2}_{b,c,k,s}, \tag{31}$$

$$\frac{\partial \mathcal{L}}{\partial B_{c,k}} = \sum_{b=1}^{B}\sum_{s=1}^{S}\left(\sum_{o=1}^{O}\Delta Y_{b,o,s}W^{q}_{o,c,k}\right)A_{c,k}\widetilde{P}^{2}_{b,c,k,s}. \tag{32}$$

The gradient of the filtered local feature $\widetilde{P}_{b,c,k,s}$ is

$$\frac{\partial \mathcal{L}}{\partial \widetilde{P}_{b,c,k,s}} = 2\left(\sum_{o=1}^{O}\Delta Y_{b,o,s}W^{q}_{o,c,k}\right)A_{c,k}B_{c,k}\widetilde{P}_{b,c,k,s}. \tag{33}$$

The above formula shows that the backpropagation of the quadratic term in STKQC can be viewed as the output gradient ΔY being propagated backward through the quadratic projection weights W^{q} to the quadratic response R, and then further applied to the learnable quadratic mapping parameters A, B, and the local features $\widetilde{P}$ selected via soft top-k. Since the quadratic response adopts a diagonal quadratic form, its gradient is relatively simple and is proportional to the current filtered local features $\widetilde{P}_{b,c,k,s}$.

Finally, the patch-space gradients from the quadratic branch are restored to the input feature map space via the corresponding backpropagation through the unfold operation, and act jointly with the gradients from the standard convolutional branch on the input features:

$$\frac{\partial \mathcal{L}}{\partial \mathbf{X}} = \left.\frac{\partial \mathcal{L}}{\partial \mathbf{X}}\right|_{\text{lin}} + \left.\frac{\partial \mathcal{L}}{\partial \mathbf{X}}\right|_{\text{quad}}. \tag{34}$$

Here, the gradient of the standard convolution branch is obtained via standard backpropagation, while the gradient of the quadratic branch is propagated via the aforementioned quadratic response path. Since the core operations of STKQC—including unfold, softmax, sigmoid, element-wise multiplication, matrix projection, and convolution—are all differentiable operations, this module can be optimized end-to-end using standard backpropagation.

3.4. Relation to existing second-order and bilinear methods

Existing second-order convolution methods (Jiang et al., 2020; Yu & Salzmann, 2017), bilinear pooling methods (Lin & Maji, 2017), and Volterra or polynomial convolutional neural networks typically enhance feature representations by explicitly modeling nonlinear feature interactions. Overall, these methods aim to capture pairwise relationships among elements within local feature vectors. Although such modeling approaches are expressive, applying them directly to convolutional feature maps results in high computational costs. For convolutional kernels with large spatial dimensions or a high number of input channels, the corresponding local feature representations become high-dimensional. Therefore, as the channel dimension or kernel size increases, fully modeling all pairwise interactions between spatial and channel elements leads to a sharp increase in the number of parameters and computational cost.

Volterra or polynomial-based convolutional neural networks (Li et al., 2022) can be viewed as a broader class of higher-order nonlinear filtering methods. They typically extend standard linear convolution by introducing second-order or higher-order polynomial terms, enabling the network to approximate more complex nonlinear mappings. However, such methods often rely on explicit higher-order kernel functions or predefined polynomial expansions. When applied to convolutional feature maps, these formulas may still require a large number of parameters, especially when considering dense inter-channel and inter-spatial interactions. Furthermore, many polynomial convolutional methods focus on improving the nonlinear approximation capability of the convolutional operator itself, rather than designing adaptive mechanisms to selectively preserve information-rich local responses in dense prediction tasks.

The proposed STKQC differs from existing fully quadratic, bilinear, Volterra-based, and polynomial convolution formulas in three aspects. First, STKQC does not explicitly construct a quadratic interaction space across all spatio-channel elements in the receptive field; instead, it utilizes lightweight learnable projections to generate a compact quadratic representation. This avoids the direct parameterization of all pairwise interactions and makes the module easier to integrate into standard convolutional segmentation networks.

Second, STKQC introduces a differentiable soft k-selection mechanism. Unlike methods that treat all elements in the receptive field equally, this module adaptively emphasizes informative local responses through spatial selection factors and dimensionality selection mappings. This distinguishes it from other approaches. In this way, STKQC aims to preserve useful local second-order information while suppressing redundant or low-information responses.

Third, STKQC does not aim to completely replace the entire network with high-order representations, but rather serves as a plug-in enhancement to standard convolutional layers. By combining STKQC's output with the original linear convolutional response, the network retains the stable representation capabilities of linear convolution while incorporating additional nonlinear local interactions. This distinguishes STKQC from global second-order pooling methods, which are typically designed for image-level representations, as well as from general polynomial convolutional networks that primarily emphasize high-order function approximation. STKQC is better suited for dense prediction tasks such as medical image segmentation, where it is essential to simultaneously preserve spatial resolution, local boundary information, and computational efficiency.

In summary, while previous second-order, bilinear, Volterra-based, and polynomial convolution methods have demonstrated advantages in modeling nonlinear or pairwise feature interactions, STKQC focuses on providing a lightweight, locally adaptive, and differentiable formulation for volumetric segmentation networks. Its main contribution lies not only in the introduction of quadratic terms but also in providing an efficient soft selection strategy that achieves local second-order feature enhancement with controllable computational overhead.

4. Experimental analysis

In this experiment, we evaluate our proposed STKQC on medical image segmentation tasks. We incorporate several standard networks and replace some of their convolutions with our STKQC to determine whether it can improve the networks' inference performance. This experiment was conducted using two NVIDIA RTX 3090 Ti GPUs paired with an Intel i9-13900K CPU.

In the segmentation task, we first selected the cardiac segmentation dataset used in MICCAI 2017—the Automated Cardiac Diagnosis Challenge (ACDC) (Bernard et al., 2018)—as the test dataset. Currently, this dataset has made the complete training data and test data publicly available, along with the evaluation scripts on its official website. Therefore, we can conveniently test the network performance and provide convincing evaluation metrics. This dataset was created by the University Hospital of Dijon, which obtained real clinical examination results, processed them in accordance with local ethics committee regulations, and anonymized them. The dataset includes 150 subjects divided into five subgroups: normal cases (NOR); previous myocardial infarction (MINF); dilated cardiomyopathy (DCM); hypertrophic cardiomyopathy (HCM); and right ventricular abnormality (ARV). MRI scans from 100 participants in the dataset are used as training data, covering both the end-diastolic and end-systolic phases. MRI scans from the remaining 50 participants are used as a test set.

Abdominal multi-organ segmentation is one of the most attractive research directions in the field of medical image analysis. In recent years, deep learning techniques have achieved significant success in abdominal multi-organ segmentation, but they have also exposed the lack of large-scale comprehensive benchmark datasets. Currently, there

Table 1
Metrics for networks on the ACDC test set.

Methods	LV		Myo		RV	
	DSC	HD	DSC	HD	DSC	HD
STKQC_BaseNet_N	91.82 ± 6.84	9.84 ± 9.24	86.49 ± 3.72	11.26 ± 9.42	88.03 ± 7.13	14.32 ± 8.81
STKQC_BaseNet_E	91.80 ± 6.69	8.85 ± 5.22	85.68 ± 3.49	9.93 ± 4.80	87.27 ± 8.26	15.14 ± 6.11
STKQC_BaseNet_B	93.53 ± 5.32	7.63 ± 4.20	88.32 ± 3.17	9.29 ± 4.05	89.16 ± 7.03	13.27 ± 5.83
STKQC_BaseNet_D	92.93 ± 5.25	8.55 ± 4.29	87.22 ± 3.54	10.68 ± 4.94	87.93 ± 7.44	14.60 ± 7.46

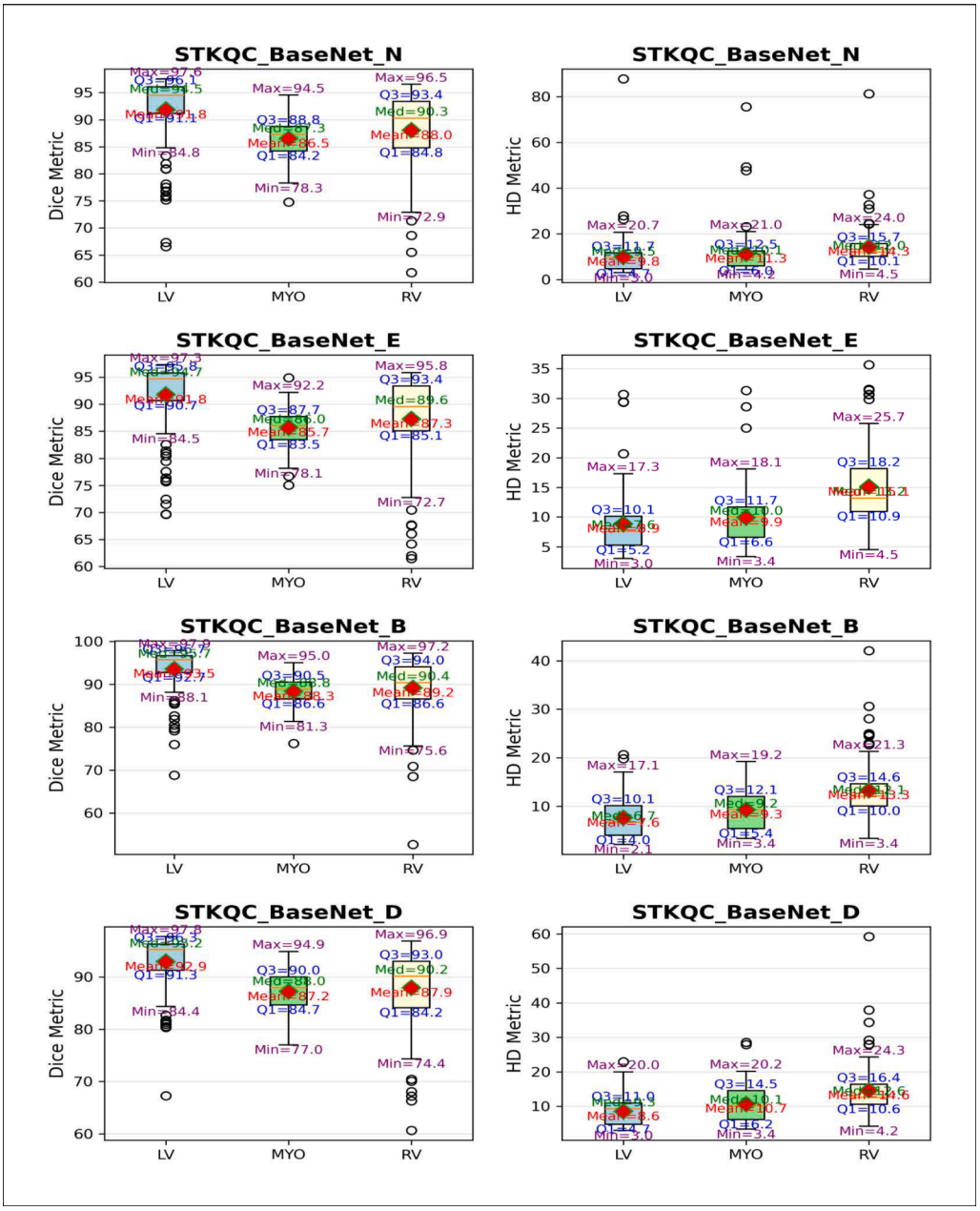


Fig. 2. details for networks on the ACDC test set.

Table 2
DSC on the ACDC test set.

Methods	LV		Myo		RV	
	DSC	HD	DSC	HD	DSC	HD
UNet	91.87 ± 6.30	9.38 ± 6.19	86.44 ± 3.77	11.30 ± 7.52	88.01 ± 8.08	14.78 ± 8.44
TKQC-UNet	92.61 ± 5.50	8.63 ± 5.53	86.13 ± 3.88	10.57 ± 5.42	88.59 ± 7.36	14.85 ± 7.13
STKQC-UNet	93.03 ± 5.71	8.02 ± 4.64	88.03 ± 3.28	9.63 ± 4.71	88.73 ± 7.51	14.15 ± 7.85

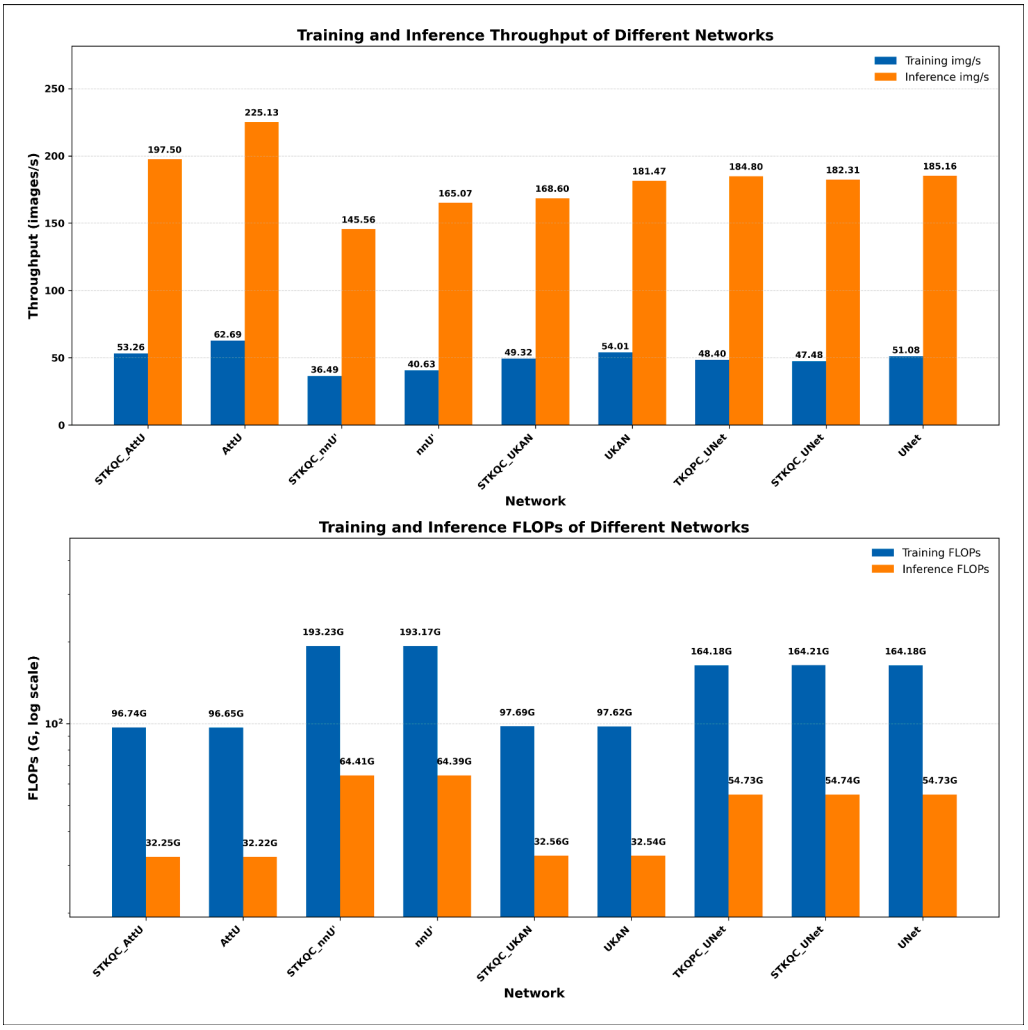


Fig. 3. Comparison of computational efficiency between the baseline network and the STKQC-enhanced network.

Table 3
DSC of different models on the ACDC test set.

Methods	LV		Myo		RV	
	ED	ES	ED	ES	ED	ES
UNet	95.30 ± 1.78	88.45 ± 7.27	85.70 ± 4.04	87.19 ± 3.33	91.76 ± 4.78	84.27 ± 8.94
STKQC-UNet	95.65 ± 2.02	90.41 ± 6.96	87.03 ± 3.09	88.20 ± 3.15	91.90 ± 5.42	85.58 ± 8.08
Att-UNet	95.04 ± 1.90	89.63 ± 5.86	84.59 ± 4.38	87.14 ± 3.56	91.36 ± 5.40	84.38 ± 8.71
STKQC-AttU	95.95 ± 1.50	90.59 ± 6.63	87.11 ± 3.45	88.99 ± 3.24	91.07 ± 5.83	86.34 ± 9.09
UKAN	95.63 ± 1.92	88.60 ± 7.93	86.64 ± 3.13	87.65 ± 3.11	92.79 ± 3.79	84.76 ± 10.58
STKQC-UKAN	95.73 ± 1.48	90.27 ± 7.02	86.53 ± 3.91	88.04 ± 3.69	92.33 ± 4.35	85.27 ± 10.52
nnU'	95.61 ± 1.85	89.18 ± 8.45	86.34 ± 2.98	87.49 ± 3.24	91.13 ± 5.78	82.54 ± 13.25
STKQC-nnU'	95.82 ± 1.63	88.85 ± 8.87	86.63 ± 3.21	87.82 ± 2.91	92.21 ± 4.41	84.56 ± 9.15

are several benchmark datasets for abdominal organ segmentation, but they suffer from insufficient organ numbers and training samples, making it difficult to comprehensively and fairly evaluate different methods. Therefore, the AMOS22 (Ji et al., 2022) abdominal multi-organ segmentation benchmark dataset was proposed. This dataset includes scan data from multi-center, multi-device, multi-modal, multi-phase, and multi-disease patients. Each dataset contains voxel-level annotations for 15 abdominal organs, including the spleen, right kidney, left kidney, gallbladder, esophagus, liver, stomach, aorta, inferior vena cava, pancreas, right adrenal gland, left adrenal gland, duodenum, bladder, and prostate/uterus. In this experiment, we utilized the publicly available portion of the dataset, comprising 240 patient datasets for training (200 CT, 40 MRI) and 120 patient datasets for validation and testing (100 CT, 20 MRI).

We performed preprocessing on the acquired datasets. All data were first resampled; the resampled pixdim for the ACDC dataset was $1.5mm \times 1.5mm \times 10mm$, and for the AMOS dataset, it was $1.0mm \times 1.0mm \times 5.0mm$. Subsequently, we applied padding to both datasets, resizing the ACDC dataset to 384×384 pixels and the AMOS dataset to 580×580 pixels. Finally, we expanded both datasets for training through affine and elastic transformations. During network training, the data from all 100 ACDC participants and their corresponding labels were divided into several

Table 4
HD of different models on the ACDC test set.

Methods	LV		Myo		RV	
	ED	ES	ED	ES	ED	ES
UNet	7.56 ± 5.24	11.21 ± 6.51	9.93 ± 7.33	12.67 ± 7.51	12.60 ± 4.91	16.95 ± 10.45
STKQC-UNet	6.71 ± 3.68	9.34 ± 5.18	8.50 ± 4.71	10.76 ± 4.53	12.71 ± 6.89	15.59 ± 8.61
Att-UNet	8.23 ± 4.84	11.19 ± 6.05	10.19 ± 4.38	11.74 ± 5.75	12.64 ± 4.94	15.49 ± 3.55
STKQC-AttU	6.65 ± 1.50	9.61 ± 6.63	8.45 ± 4.23	10.47 ± 4.22	13.75 ± 10.69	15.33 ± 6.98
UKAN	7.37 ± 6.00	12.73 ± 7.48	10.47 ± 4.38	15.96 ± 8.28	13.57 ± 11.65	17.43 ± 9.13
STKQC-UKAN	7.14 ± 3.11	9.66 ± 3.80	9.36 ± 4.38	11.77 ± 4.92	13.38 ± 4.50	16.29 ± 8.75
nnU′	7.99 ± 5.16	12.33 ± 11.02	10.57 ± 6.16	13.09 ± 7.29	13.59 ± 6.12	16.96 ± 8.16
STKQC-nnU′	7.69 ± 4.03	13.57 ± 15.33	10.70 ± 9.88	13.19 ± 13.44	13.96 ± 6.82	15.93 ± 6.81

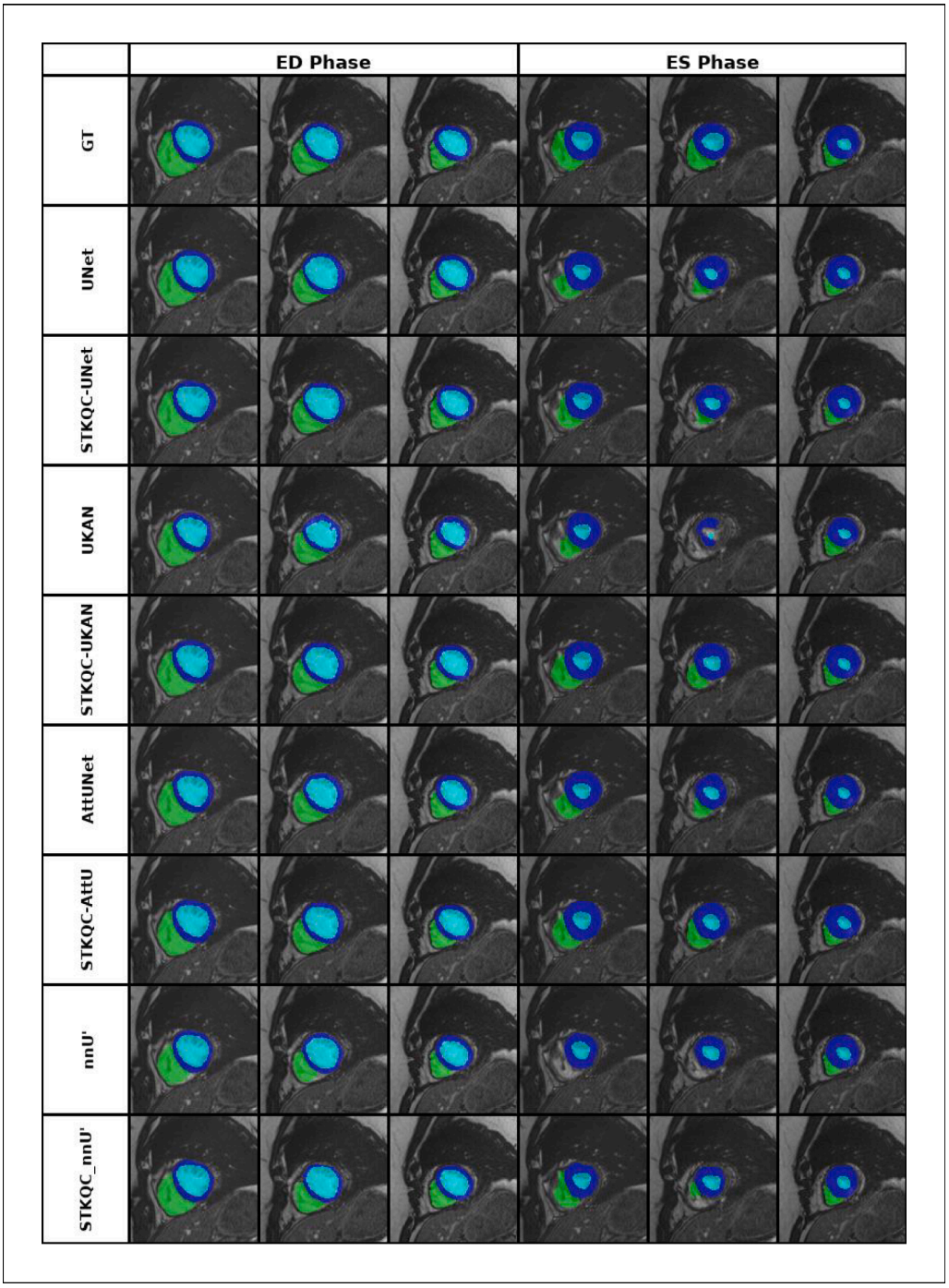


Fig. 4. The predictions of ACDC images using the baseline network and the corresponding network introducing STKQC.

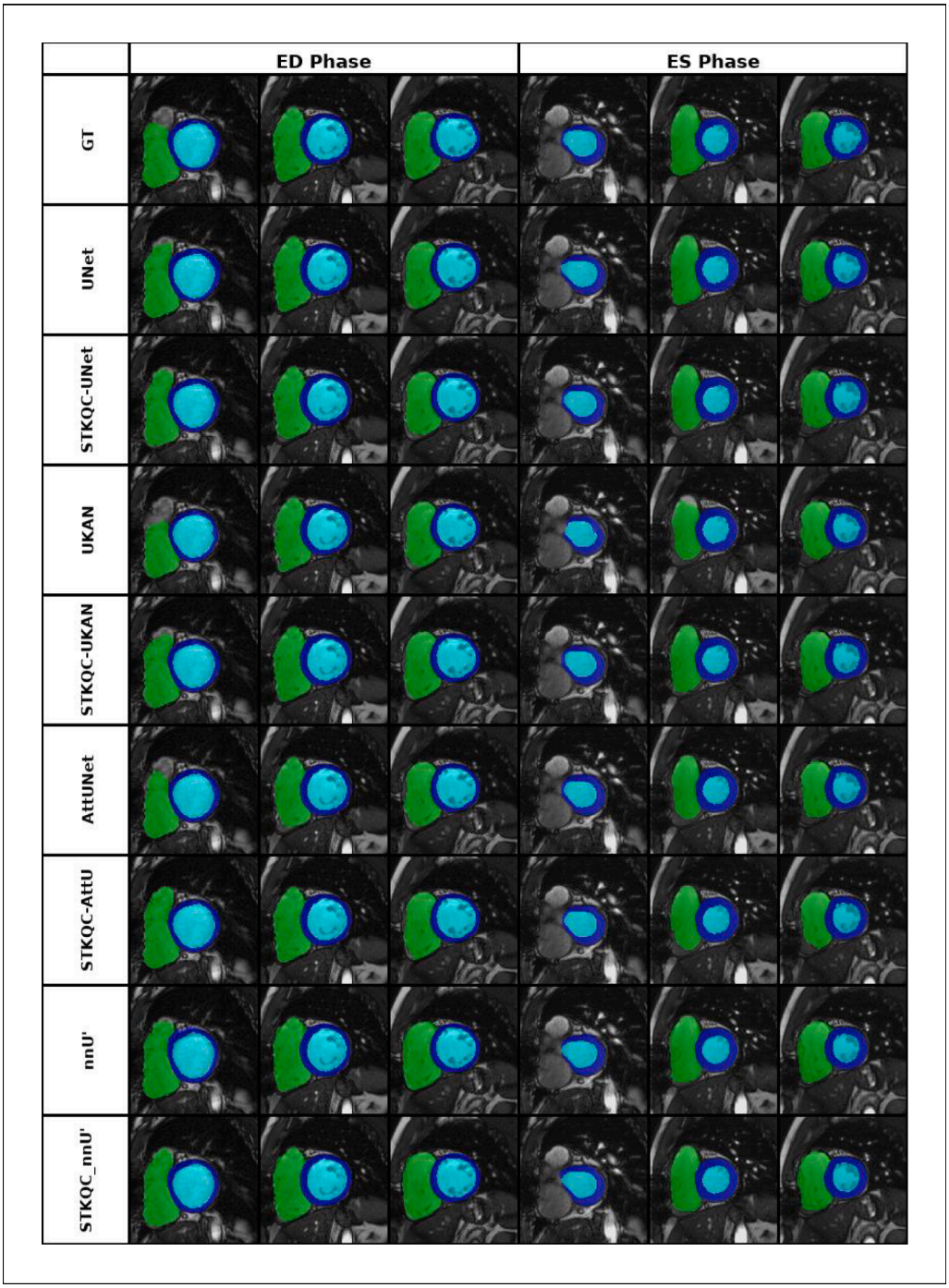


Fig. 5. The predictions of ACDC images using the baseline network and the corresponding network introducing STKQC.

Table 5
Statistical significance analysis between baseline models and STKQC-enhanced models on the ACDC test set.

Comparison	DSC Diff. ↑	DSC p-value	HD Diff. ↓	HD p-value
UNet & STKQC-UNet	+1.154	< 0.001	−1.220	0.033
Att-UNet & STKQC-AttU	+1.315	< 0.001	−0.871	0.012
UKAN & STKQC-UKAN	+0.343	0.102	−1.652	0.114
nnU′ & STKQC-nnU′	+0.597	0.026	+0.088	0.310

two-dimensional slices. Following the instructions, the data was divided into five groups. For each group, the first 70% was combined to form the training set, and the remaining 30% was combined to form the test set. The data from the remaining 50 participants was used as the test set. For the AMOS dataset, the publicly available data from 240 participants was used for training. The data from the first 60 participants in the validation set was used as the validation set, and the remaining data was used as the test set. When training the AMOS dataset, we employed a batch training method. The training set was divided into 20 mutually exclusive data batches. Each batch contained resampled data, radial transformation data, and elastic transformation data from 12 participants. During training, each network under evaluation was trained batch by batch until the entire training set was processed. This training method effectively reduced memory requirements.

Table 6
DSC of different models on the AMOS test set.

Organ	UNet	STKQC-U	UKAN	STKQC-UKAN	Att-UNet	STKQC-AttU
spleen	93.77 ± 5.17	94.80 ± 4.13	93.91 ± 6.28	93.21 ± 6.53	93.53 ± 5.13	93.92 ± 6.94
right kidney	93.32 ± 12.49	93.84 ± 12.48	93.37 ± 12.55	93.57 ± 12.50	93.00 ± 12.96	93.66 ± 12.46
left kidney	93.58 ± 6.37	94.80 ± 4.13	95.12 ± 2.74	95.21 ± 2.66	93.80 ± 6.68	95.05 ± 2.93
gall bladder	63.17 ± 31.43	67.77 ± 30.70	64.48 ± 33.31	66.35 ± 31.60	62.67 ± 31.62	67.42 ± 31.76
esophagus	69.33 ± 20.58	74.41 ± 16.76	71.85 ± 16.44	73.38 ± 16.62	70.38 ± 17.23	71.41 ± 18.24
liver	94.68 ± 4.87	95.58 ± 4.05	95.42 ± 3.36	95.67 ± 2.88	95.34 ± 3.88	95.64 ± 3.53
stomach	83.20 ± 14.29	86.19 ± 16.81	85.76 ± 12.11	84.43 ± 16.68	83.45 ± 14.70	85.30 ± 13.63
arota	92.68 ± 3.84	93.64 ± 3.36	92.88 ± 5.31	93.45 ± 3.46	92.18 ± 5.60	92.99 ± 5.36
postcava	74.82 ± 15.02	81.34 ± 10.07	80.37 ± 9.83	78.84 ± 12.99	76.86 ± 12.59	79.68 ± 13.35
pancreas	69.67 ± 18.57	73.14 ± 15.58	71.91 ± 17.20	73.08 ± 16.33	72.39 ± 15.09	72.64 ± 15.43
right adrenal gland	58.36 ± 19.27	69.06 ± 13.56	46.02 ± 22.85	59.08 ± 19.78	59.95 ± 19.02	64.28 ± 16.61
left adrenal gland	50.50 ± 23.58	61.23 ± 20.47	55.51 ± 19.06	53.56 ± 21.52	55.15 ± 20.86	61.57 ± 17.78
duodenum	58.86 ± 18.77	66.27 ± 18.39	62.46 ± 19.46	65.08 ± 18.47	62.16 ± 18.58	63.60 ± 20.74
bladder	81.74 ± 14.73	83.51 ± 18.40	78.62 ± 20.92	81.70 ± 21.37	79.63 ± 20.80	79.94 ± 21.48
prostate/uterus	66.50 ± 18.64	76.80 ± 21.86	72.63 ± 24.65	69.49 ± 26.40	71.06 ± 23.47	74.51 ± 22.89

Table 7
HD95 of different models on the AMOS test set.

Organ	UNet	STKQC-U	UKAN	STKQC-UKAN	Att-UNet	STKQC-AttU
spleen	7.29 ± 9.31	5.57 ± 6.89	6.60 ± 10.03	6.61 ± 8.87	6.94 ± 9.04	5.69 ± 8.14
right kidney	7.57 ± 38.50	7.78 ± 38.70	6.19 ± 25.72	5.84 ± 25.65	7.59 ± 38.59	7.48 ± 38.54
left kidney	7.48 ± 21.43	5.75 ± 20.63	5.87 ± 20.62	5.74 ± 20.54	6.57 ± 20.85	5.93 ± 20.55
gall bladder	46.41 ± 88.73	37.78 ± 81.64	32.47 ± 57.93	27.86 ± 50.06	40.07 ± 82.87	41.28 ± 84.21
esophagus	24.37 ± 36.76	18.53 ± 29.51	17.84 ± 27.54	17.55 ± 29.61	20.75 ± 29.28	25.21 ± 37.92
liver	11.16 ± 16.39	9.48 ± 16.78	9.04 ± 15.15	8.09 ± 10.84	8.52 ± 14.26	8.58 ± 13.63
stomach	20.11 ± 22.87	17.08 ± 18.56	14.70 ± 14.30	16.26 ± 18.73	20.80 ± 19.88	17.35 ± 18.58
arota	10.02 ± 20.48	7.58 ± 18.12	10.07 ± 21.85	7.95 ± 18.64	11.51 ± 24.18	9.07 ± 21.47
postcava	36.03 ± 42.02	18.17 ± 26.21	17.75 ± 26.95	24.29 ± 35.01	30.44 ± 37.59	19.00 ± 30.73
pancreas	18.43 ± 19.43	16.24 ± 18.33	16.31 ± 19.20	17.00 ± 21.40	14.09 ± 17.66	15.33 ± 15.80
right adrenal gland	7.94 ± 5.99	6.12 ± 6.28	18.07 ± 35.21	10.88 ± 25.61	8.43 ± 6.62	6.81 ± 5.68
left adrenal gland	16.80 ± 38.58	9.00 ± 8.14	10.48 ± 7.69	11.08 ± 11.31	13.48 ± 18.39	9.25 ± 6.85
duodenum	30.61 ± 41.00	26.40 ± 40.85	26.79 ± 29.76	23.30 ± 29.03	27.65 ± 40.64	29.57 ± 45.64
bladder	11.76 ± 9.36	10.97 ± 13.73	13.27 ± 12.46	12.89 ± 16.70	13.16 ± 16.18	16.70 ± 39.11
prostate/uterus	18.31 ± 39.08	9.98 ± 10.01	13.09 ± 26.32	17.70 ± 35.74	15.58 ± 38.52	15.35 ± 38.44

Table 8
Statistical significance analysis between baseline models and STKQC-enhanced models on the AMOS test set.

Comparison	DSC Diff. ↑	DSC p-value	HD95 Diff. ↓	HD95 p-value
UNet & STKQC-UNet	+4.068	< 0.001	−3.680	< 0.001
Att-UNet & STKQC-AttU	+2.003	< 0.001	−0.818	0.053
UKAN & STKQC-UKAN	+1.009	0.017	−0.368	0.643

In the experiments, we use ADAM as the optimizer and cross-entropy loss as the loss function. On the ACDC dataset, we use the Dice Similarity Coefficient (DSC) and Hausdorff distance (HD) as evaluation metrics, while on the AMOS dataset, we use DSC and the 95% Hausdorff distance (HD95) as evaluation metrics. In the ACDC dataset, all networks started with an initial learning rate of $1e-3$ and were trained for 60 epochs; after 30 epochs of training, the learning rate was reduced to $1e-4$. In the AMOS dataset, all networks also started with an initial learning rate of $1e-3$ and were trained for 180 epochs. After every 60 epochs of training, the learning rate was reduced to one-tenth of its previous value.

STKQC's position-dependent performance on model inference was first evaluated; given the symmetry of the U-shaped network architecture, we introduced a U-shaped network for testing. Specifically, we refer to the network where convolutions in the encoding path are replaced with STKQC as STKQC_BaseNet_E, the network where the decoding path is replaced as STKQC_BaseNet_D, and the network where the bottleneck layer is replaced as STKQC_BaseNet_B. STKQC_BaseNet_N represents the network where no convolutions are replaced. All four networks were tested under identical conditions. Their performance on the ACDC test set is shown in the Table 1, with details illustrated in Fig. 2. The experimental results indicate that replacing convolutional operations with STKQC improves the networks' inference performance, with models incorporating a bottleneck layer demonstrating the most stable performance.

By combining adaptive factors and a modulating graph, we redistribute the weights of input features across both spatial and dimensional dimensions, thereby achieving a differentiable SoftTopK feature selection method that facilitates network training. We also tested a U-shaped network that employs a non-differentiable Top K approach for feature selection without using adaptive factors or a modulating graph. Under identical experimental conditions, the experimental results are shown in the Table 2.

To evaluate the performance of STKQC, we use Attention UNet (Schlemper et al., 2019), UKAN (Li et al., 2025), and UNet as baseline networks. Additionally, since the proposed STKQC currently operates only on conv2d, we introduce a network denoted as nnU′ based on the nnUNet (Isensee et al., 2021) architecture. nnU′ does not include operations such as automatic preprocessing found in nnUNet; it merely adopts the network architecture portion. We incorporate STKQC into the baseline networks, naming them STKQC_AttentionUNet, STKQC_UKAN, STKQC_UNet, and STKQC_nnU′, respectively.

To verify the impact of introducing STKQC on model computational efficiency, this paper compiles relevant data for various baseline networks and their STKQC-enhanced versions. In the experiments, all networks were evaluated using a uniform input size (256×256), the same number of classes, and the same testing environment. Training FLOPs are used to reflect the computational requirements of the model during forward propagation, backward propagation, and parameter updates, while inference FLOPs are used to measure the computational overhead during the actual deployment phase. Additionally, images processed

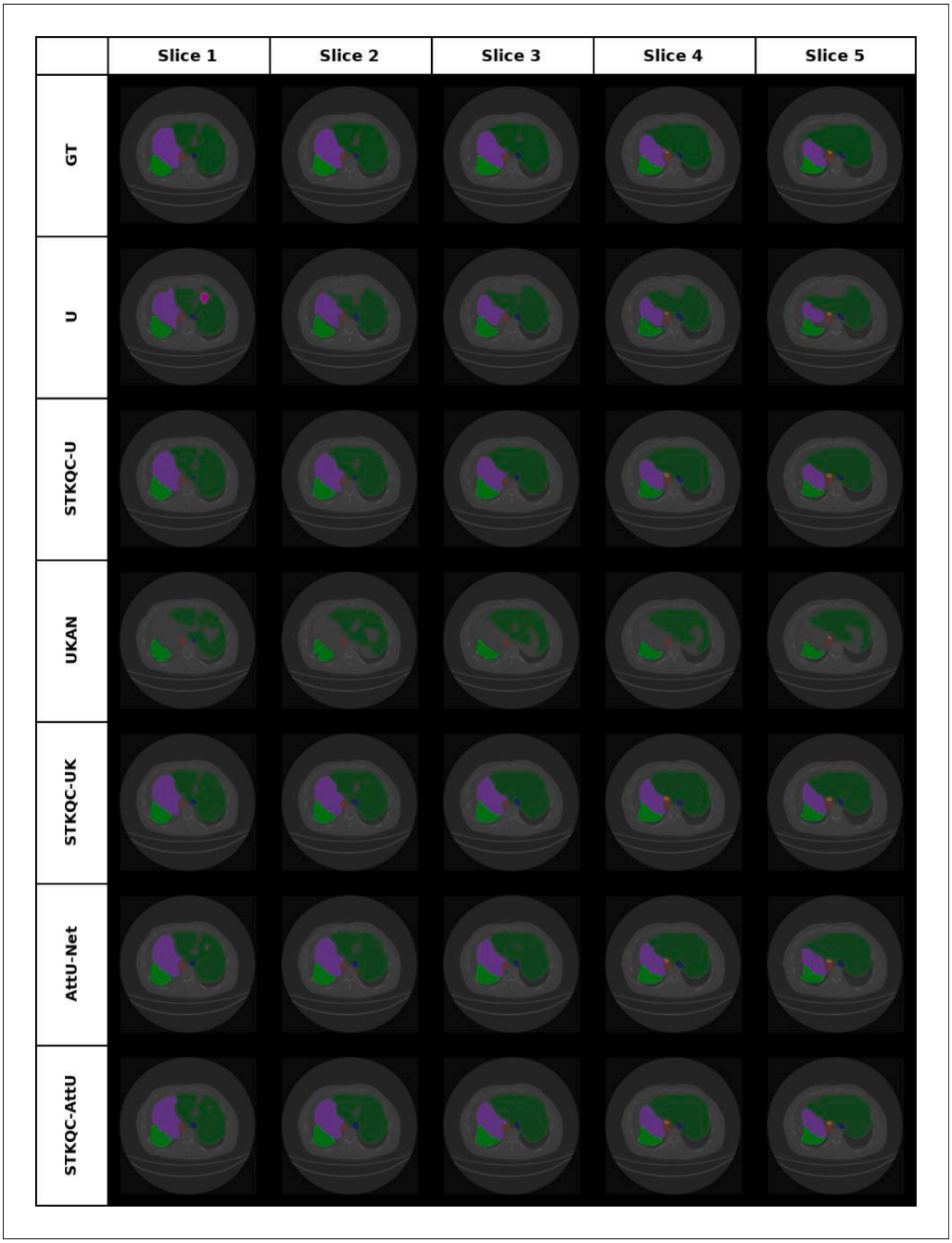


Fig. 6. The predictions of AMOS images using the baseline network and the corresponding network introducing STKQC.

per second during training and inference further validate the model's efficiency in terms of processing speed. By comparing the FLOPs and throughput of the original network with its corresponding STKQC version, the additional computational cost introduced by this module can be intuitively assessed. Fig. 3 illustrates that although STKQC introduces quadratic feature modeling capabilities on top of standard convolutions, its use of the Soft TopK selection mechanism—which models quadratic interactions only for key feature dimensions within local responses—means it does not significantly increase the overall computational burden.

In this experiment, we replace the linear convolutions in the bottleneck layers of the baseline network with STKQC. We evaluated the impact of STKQC on the model's inference performance using two datasets. After training under identical experimental conditions, the network's performance on the test sets of both datasets, along with selected inference performance metrics, is shown in the table and figure below.

Tables 3 and 4 show the DSC and HD results of different networks on the ACDC dataset, respectively. As shown in the tables, under the same experimental conditions, the improved networks incorporating STKQC outperform their corresponding baseline networks to some extent. Furthermore, Figs. 4 and 5 compare the image prediction accuracy of the baseline networks and their corresponding improved versions in the ventricular segmentation task. For images with simple shape patterns, all networks provide accurate predictions. However, for images with complex features (such as blurred contours), the networks enhanced with STKQC still provide more accurate predictions compared to the baseline networks.

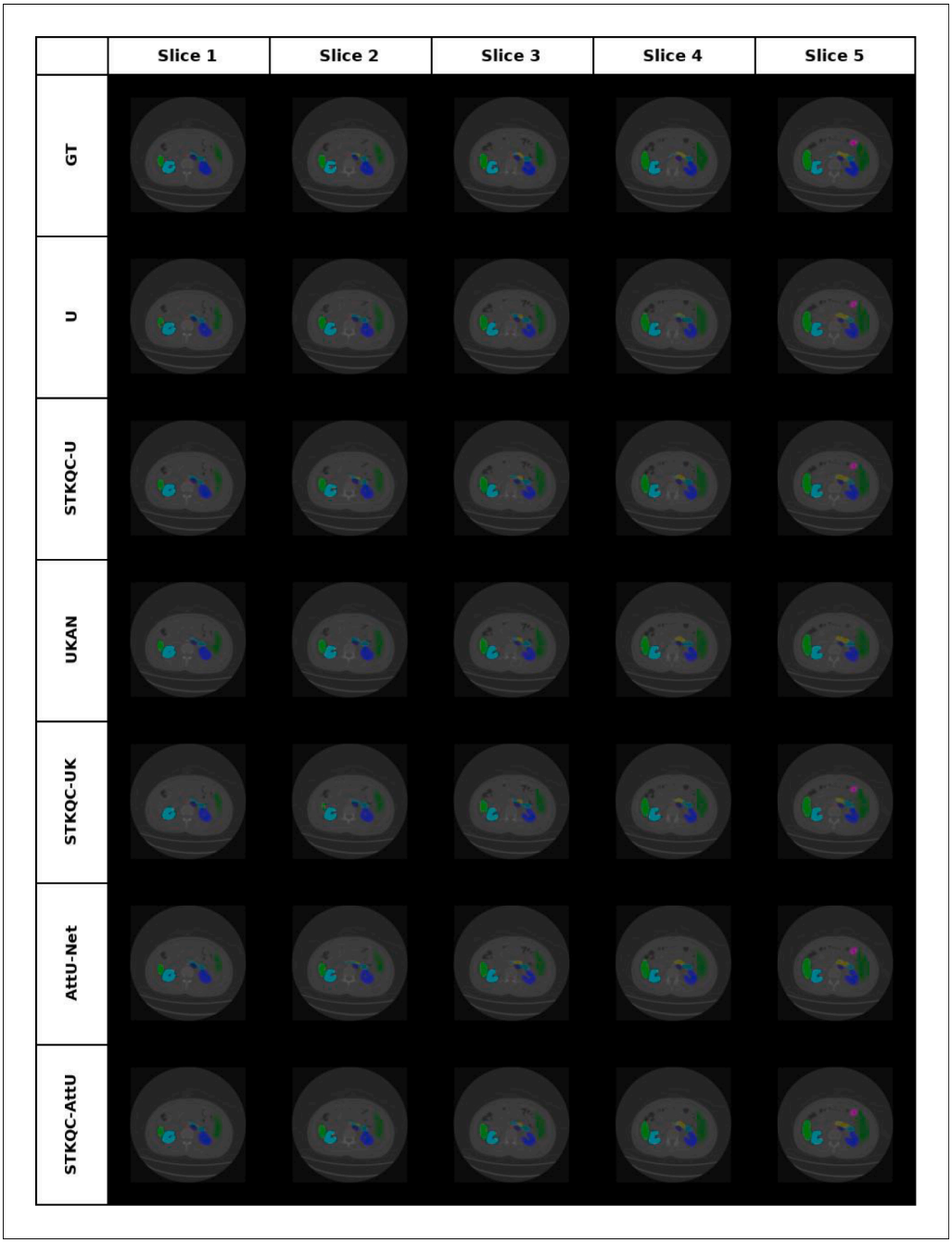


Fig. 7. The predictions of AMOS images using the baseline network and the corresponding network introducing STKQC.

Tables 6 and 7 present a comparison of image prediction accuracy between the baseline network and the corresponding improved networks on the AMOS task. The data in the tables clearly demonstrate that, under identical conditions, the improved networks exhibit enhanced performance in organ prediction compared to the original network. Furthermore, in the vast majority of cases, the HD95 score also performs better. Figs. 6 and 7 display the mask prediction results for each network. The images show that, in most cases, the prediction results of the improved network are similar to those of the baseline network; however, for the small organs shown in the figures, the improved network is also capable of making more accurate predictions compared to the baseline network.

To further assess whether the performance improvements are consistent at the individual case level, we conducted paired Wilcoxon signed-rank tests between each base model and its corresponding STKQC-enhanced version, using case-level ED and ES results from the ACDC test set. As shown in Table 5, STKQC yielded statistically significant improvements in mean DSC for UNet, Att-UNet, and nnU′. For UKAN, STKQC also increased the mean DSC, but the difference was not statistically significant. Regarding the HD metric, the introduction of STKQC significantly reduced HD for UNet and Att-UNet, while the difference in HD was not statistically significant on UKAN and nnU′.

As shown in the Table 8, significance tests conducted on the AMOS22 dataset reveal that STKQC significantly improves the Mean DSC across all tested baseline networks, indicating that it consistently enhances the accuracy of region overlap in multi-organ segmentation. For HD95, STKQC achieves significant improvements on UNet, approaches significance on Att-UNet, and fails to reach significance on UKAN, suggesting

that its impact on boundary distance metrics is dependent on the underlying network architecture.

5. Discussion

In this study, we designed experiments to compare whether the introduction of STKQC can improve the performance of baseline networks in medical image segmentation tasks. The results show that introducing STKQC enhances the feature representation capabilities of the segmentation network. Compared to standard convolutional layers, STKQC introduces additional local quadratic interactions, enabling the network to capture more complex nonlinear relationships within the local receptive field. We also tested the impact of introducing STKQC at different locations on model performance. In an encoder-decoder architecture, shallow layers are responsible for capturing low-level details such as edges, textures, and local brightness changes, while deep and bottleneck layers are responsible for extracting high-level semantic representations. The decoder path progressively restores spatial resolution. Although STKQC is designed as a lightweight quadratic convolution module, it still introduces additional learnable parameters and nonlinear operations compared to standard convolution layers. STKQC should be regarded as a plug-in module that requires appropriate placement and configuration, rather than a component to be inserted as frequently as possible. Future research could further explore optimal insertion strategies for STKQC. Additionally, adaptive strategies could be investigated to automatically determine the optimal insertion points for STKQC based on the characteristics of the backbone network and the target segmentation task. Alternatively, its underlying principles could be applied to upsampling algorithms, thereby opening up further possibilities. Furthermore, although this study focuses on medical image segmentation, STKQC's local quadratic interaction mechanism does not rely on prior knowledge specific to medical images. Therefore, from a methodological perspective, it may also be applicable to non-medical scenarios such as natural image segmentation, remote sensing image segmentation, and industrial vision. However, this paper has not yet conducted experimental validation in these scenarios, and its effectiveness and cross-domain generalization capabilities still need to be confirmed through further systematic research. Overall, the experimental results demonstrate that STKQC effectively improves the performance of medical image segmentation networks by enhancing the modeling of local nonlinear features.

6. Conclusion

This paper proposes a novel soft-top k-quadratic convolutional module (STKQC) designed to enhance the nonlinear feature representation capabilities of standard convolutional networks. By introducing a lightweight quadratic branch, STKQC enables the network to capture local second-order feature interactions that are difficult to model using purely linear convolutions. Furthermore, the proposed soft-top-k selection strategy adaptively emphasizes information-rich local responses while suppressing redundant components, thereby enhancing the effectiveness and stability of the quadratic representation.

Experimental results on medical image segmentation tasks demonstrate that integrating STKQC into various baseline networks consistently improves segmentation performance while maintaining a controllable computational cost. These results indicate that STKQC can serve as a flexible plug-in module to enhance convolutional segmentation networks, particularly in scenarios where local structural details and nonlinear feature interactions are critical.

In future research, we will further explore the application potential of quadratic convolutions in more complex medical image analysis scenarios. Specifically, we plan to extend STKQC to 3D convolutional architectures to enable volumetric image segmentation. Additionally, we will investigate combining quadratic feature modeling with deconvolution or upsampling operations to improve feature reconstruction and boundary refinement in dense prediction tasks

List of abbreviations

The abbreviations used in this manuscript are summarized in the table below(Table 9).

Table 9
List of Abbreviations.

Acronym	Full name
CNN	Convolutional Neural Network
STKQC	Soft TopK Quadratic Convolution
UNet	U-shaped Network
UKAN	U-Net with Kolmogorov–Arnold Networks
nnU-Net	nnU-Net framework
nnU′	nnU-Net like framework
ACDC	Automated Cardiac Diagnosis Challenge
AMOS	Abdominal Multi-Organ Segmentation
DSC	Dice Similarity Coefficient
HD	Hausdorff Distance
HD95	95th percentile Hausdorff Distance
FLOPs	Floating Point Operations
FPS	Frames Per Second
ED	End-Diastole
ES	End-Systole
GT	Ground Truth

CRediT authorship contribution statement

Hongyang Zhang: Writing – original draft, Validation, Project administration, Formal analysis, Data curation, Conceptualization; **Keqiang Wang:** Writing – original draft, Visualization, Methodology, Data curation; **Jie Chen:** Visualization, Resources, Conceptualization; **Peng Wang:** Supervision, Resources, Project administration, Investigation; **Yuanquan Wang:** Writing – review & editing, Validation, Supervision, Methodology, Funding acquisition, Formal analysis, Conceptualization; **Wenxue Zhang:** Writing – review & editing, Supervision, Resources, Project administration, Funding acquisition, Formal analysis, Data curation, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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References

Abdeltawab, H., Khalifa, F., Taher, F., Alghamdi, N. S., Ghazal, M., Beache, G., Mohamed, T., Keynton, R., & El-Baz, A., (2020). A deep learning-based approach for automatic segmentation and quantification of the left ventricle from cardiac cine MR images. *Computerized Medical Imaging and Graphics*, *81*, 101717.

Bernard, O., Lalande, A., Zotti, C., Cervenansky, F., Yang, X., Heng, P.-A, Cetin, I., Lekadir, K., Camara, O., Gonzalez Ballester, M. A., Sanroma, G., Napel, S., Petersen, S., Tziritas, G., Grinias, E., Khened, M., Kollerathu, V. A., Krishnamurthi, G., Rohé, M.-M. …, Pennec, X., (2018). Deep learning techniques for automatic MRI cardiac multi-structures segmentation and diagnosis: Is the problem solved? *IEEE Transactions on Medical Imaging*, *37* (11), 2514–2525.

Boykov, Y., & Jolly, M.-P. (2001). Interactive graph cuts for optimal boundary & region segmentation of objects in n-d images. In *Proceedings of the eighth IEEE international conference on computer vision (ICCV 2001)* (pp. 105–112). (*vol. 1*).

Cai, Z., Xin, J., Shi, P., Wu, J., & Zheng, N. (2022). DSTUNet: UNet with efficient dense swin transformer pathway for medical image segmentation. In *2022 IEEE 19th International symposium on biomedical imaging (ISBI)* (pp. 1–5).

Cao, H., Wang, Y., Chen, J., Jiang, D., Zhang, X., Tian, X., & Wang, M. (2023). Swin-unet: Unet-like pure transformer for medical image segmentation. In *Computer vision – ECCV 2022 workshops* (pp. 205–218). Cham: Springer Nature Switzerland (*vol. 13803*). Lecture Notes in Computer Science.

Deng, J., Dong, W., Socher, R., Li, L.-J., Li, K., & Fei-Fei, L., (2009). ImageNet: A large-scale hierarchical image database. In *2009 IEEE Conference on computer vision and pattern recognition* (pp. 248–255).

Dosovitskiy, A., Beyer, L., Kolesnikov, A., Weissenborn, D., Zhai, X., Unterthiner, T., Dehghani, M., Minderer, M., Heigold, G., Gelly, S., Uszkoreit, J., & Houlsby, N. (2021). An image is worth 16x16 words: Transformers for image recognition at scale. In *International conference on learning representations (ICLR)*.

Frid-Adar, M., Diamant, I., Klang, E., Amitai, M., Goldberger, J., & Greenspan, H., (2018). GAN-based synthetic medical image augmentation for increased CNN performance in liver lesion classification. *Neurocomputing*, *321*, 321–331.

Gao, Z., Xie, J., Wang, Q., & Li, P. (2019). Global second-order pooling convolutional networks. In *Proceedings of the IEEE/CVF Conference on computer vision and pattern recognition (CVPR)* (pp. 3024–3033).

Goodfellow, I. J., Pouget-Abadie, J., Mirza, M., Xu, B., Warde-Farley, D., Ozair, S., Courville, A., & Bengio, Y., (2014). Generative adversarial nets. In*Advances in neural information processing systems* (pp. 2672–2680). (*vol. 27*).

Grau, V., Mewes, A. U. J., Alcañiz, M., Kikinis, R., & Warfield, S. K. (2004). Improved watershed transform for medical image segmentation using prior information. *IEEE Transactions on Medical Imaging*, *23* (4), 447–458.

Grzeszczyk, M. K., Płotka, S., & Sitek, A., (2022). Multi-task swin transformer for motion artifacts classification and cardiac magnetic resonance image segmentation. In *Statistical atlases and computational models of the heart. regular and CMRxmotion challenge papers* (pp. 409–417). Cham: Springer Nature Switzerland.

Han, C., Hayashi, H., Rundo, L., Araki, R., Shimoda, W., Muramatsu, S., Furukawa, Y., Mauri, G., & Nakayama, H., (2018). Gan-based synthetic brain mr image generation. In *2018 IEEE 15th International symposium on biomedical imaging (ISBI 2018)* (pp. 734–738).

He, K., Zhang, X., Ren, S., & Sun, J. (2016). Deep residual learning for image recognition. In *Proceedings of the IEEE conference on computer vision and pattern recognition (CVPR)* (pp. 770–778).

Isensee, F., Jaeger, P. F., Kohl, S. A. A., Petersen, J., & Maier-Hein, K. H. (2021). NnU-Net: A self-configuring method for deep learning-based biomedical image segmentation. *Nature Methods*, *18* (2), 203–211.

Jethi, A. K., Murugesan, B., Ram, K., & Sivaprakasam, M. (2020). Dual-encoder-u-net for fast MRI reconstruction. In *2020 IEEE 17th International symposium on biomedical imaging workshops (ISBI workshops)* (pp. 1–4).

Ji, Y., Bai, H., Ge, C., Yang, J., Zhu, Y., Zhang, R., Li, Z., Zhang, L., Ma, W., Wan, X., & Luo, P. (2022). AMOS: A large-scale abdominal multi-organ benchmark for versatile medical image segmentation. In *Advances in neural information processing systems* (pp. 36722–36732). (*vol. 35*).

Jiang, Y., Yang, F., Zhu, H., Zhou, D., & Zeng, X. (2020). Nonlinear CNN: Improving CNNs with quadratic convolutions. *Neural Computing and Applications*, *32* (12), 8507–8516.

Khened, M., Alex, V., & Krishnamurthi, G. (2018). Densely connected fully convolutional network for short-axis cardiac cine MR image segmentation and heart diagnosis using random forest. In *Statistical atlases and computational models of the heart. ACDC and MMWHS challenges* (pp. 140–151). Cham: Springer International Publishing.

Krizhevsky, A., Sutskever, I., & Hinton, G. E. (2012). Imagenet classification with deep convolutional neural networks. In *Advances in neural information processing systems* (pp. 1097–1105). Curran Associates, Inc. (*vol. 25*).

LeCun, Y., Bengio, Y., & Hinton, G. (2015). Deep learning. *Nature*, *521* (7553), 436–444.

Li, C., Liu, X., Li, W., Wang, C., Liu, H., Liu, Y., Chen, Z., & Yuan, Y., (2025). U-KAN makes strong backbone for medical image segmentation and generation. *Proceedings of the AAAI Conference on Artificial Intelligence*, *39* (5), 4652–4660.

Li, P., Xie, J., Wang, Q., & Zuo, W. (2017). Is second-order information helpful for large-scale visual recognition? In *Proceedings of the IEEE international conference on computer vision (ICCV)* (pp. 2070–2078).

Li, T., Zhou, G., Qiu, Y., & Zhao, Q. (2022). Toward understanding convolutional neural networks from volterra convolution perspective. *Journal of Machine Learning Research*, *23* (311), 1–50.

Lin, T.-Y., & Maji, S. (2017). Improved bilinear pooling with CNNs. In *Proceedings of the british machine vision conference (BMVC)* (pp. 117.1–117.12). BMVA Press.

Liu, S., Zhou, S., Wang, Y., Lu, K., Liu, W., & Wang, Z., (2025). Uncertainty-guided weakly supervised segmentation of cardiac substructures with adapter fine-tuning and fourier feature extraction. *Expert Systems with Applications*, *283*, 127583.

Liu, Y., Captur, G., Moon, J. C., Guo, S., Yang, X., Zhang, S., & Li, C., (2016). Distance regularized two level sets for segmentation of left and right ventricles from cine-MRI. *Magnetic Resonance Imaging*, *34* (5), 699–706.

Liu, Z., Lin, Y., Cao, Y., Hu, H., Wei, Y., Zhang, Z., Lin, S., & Guo, B., (2021). Swin transformer: Hierarchical vision transformer using shifted windows. In *Proceedings of the IEEE/CVF International conference on computer vision (ICCV)* (pp. 10012–10022).

Pereira, S., Pinto, A., Oliveira, J., Mendrik, A. M., Correia, J. H., & Silva, C. A., (2016). Automatic brain tissue segmentation in MR images using random forests and conditional random fields. *Journal of Neuroscience Methods*, *270*, 111–123.

Prevedello, L. M., Halabi, S. S., Shih, G., Wu, C. C., Kohli, M. D., Chokshi, F. H., Erickson, B. J., Kalpathy-Cramer, J., Andriole, K. P., & Flanders, A. E., (2019). Challenges related to artificial intelligence research in medical imaging and the importance of image analysis competitions. *Radiology: Artificial Intelligence*, *1* (1), e180031.

Ronneberger, O., Fischer, P., & Brox, T. (2015). U-Net: Convolutional networks for biomedical image segmentation. In *Medical image computing and computer-assisted intervention – MICCAI 2015* (pp. 234–241). Cham: Springer International Publishing (*vol. 9351*). Lecture Notes in Computer Science.

Schlemper, J., Oktay, O., Schaap, M., Heinrich, M., Kainz, B., Glocker, B., & Rueckert, D., (2019). Attention gated networks: Learning to leverage salient regions in medical images. *Medical Image Analysis*, *53*, 197–207.

Shelhamer, E., Long, J., & Darrell, T. (2017). Fully convolutional networks for semantic segmentation. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, *39* (4), 640–651.

Steenwijk, M. D., Pouwels, P. J. W., Daams, M., van Dalen, J. W., Caan, M. W. A., Richard, E., Barkhof, F., & Vrenken, H., (2013). Accurate white matter lesion segmentation by k nearest neighbor classification with tissue type priors (kNN-TTPs). *NeuroImage: Clinical*, *3*, 462–469.

Vaswani, A., Shazeer, N., Parmar, N., Uszkoreit, J., Jones, L., Gomez, A. N., Kaiser, Ł., & Polosukhin, I., (2017). Attention is all you need. In *Advances in neural information processing systems* (pp. 5998–6008). (*vol. 30*).

Wang, Y., Zhou, S., Lu, K., Wang, Y., Zhang, L., Liu, W., & Wang, Z., (2025). SAMBV: A fine-tuned SAM with interpolation consistency regularization for semi-supervised biventricle segmentation from cardiac mri. *Medical Engineering & Physics*, *140*, 104341.

Yu, K., & Salzmann, M. (2017). Second-order convolutional neural networks. arXiv preprint arXiv:1703.06817

Zhang, X., Zhou, S., Li, B., Wang, Y., Lu, K., Liu, W., & Wang, Z, (2025). Automatic segmentation of pericardial adipose tissue from cardiac MR images via semi-supervised method with difference-guided consistency. *Medical Physics*, *52* (3), 1679–1692.

Zhang, Z., Duan, C., Lin, T., Zhou, S., Wang, Y., & Gao, X., (2020). GVFOM: A novel external force for active contour based image segmentation. *Information Sciences*, *506*, 1–18.

Zheng, Q., Li, H., Fan, B., Wu, S., & Xu, J., (2018). Integrating support vector machine and graph cuts for medical image segmentation. *Journal of Visual Communication and Image Representation*, *55*, 157–165.